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Tutorial XII

Q1. Suppose X_1 and X_2 are random variables such that variance of each variable is 1 and the **covariance** is $\frac{1}{4}$. Determine $\text{var}\, X_1 + X_2$

Q2 Suppose that a point X is chosen in accordance with the uniform distribution on the interval $[0,1]$. Also suppose that after the value $X = x$ has been observed $(0 < x < 1)$, a point Y is chosen in accordance with uniform distribution on the interval $[x, 1]$. Determine $E[Y]$

Q3. The joint probability distribution of X and Y is

$Y \backslash X$	-1	1
0	$\frac{1}{8}$	$\frac{3}{8}$
1	$\frac{2}{8}$	$\frac{2}{8}$

Find $E[X]$, σ_X^2, $E[Y]$, σ_Y^2, $\text{Cov}(X,Y)$ and ρ_{XY}.

Q4. Let U and V be two rvs with common mean and common variance. Let $X = U+V$ and $Y = U-V$. Find $\text{Cov}(X,Y)$

Q5 Let (X,Y) be jointly distributed with density function

$$f(x,y) = \begin{cases} (x+y) & 0<x<1, \quad 0<y<1 \\ 0 & \text{otherwise} \end{cases}$$

Find $E[X^l Y^m]$, l and m are positive integers. Hence find $\text{Cov}(X,Y)$.

Tutorial XII (SOLUTION)

Q1. $Var(X_1+X_2) = Var\, X_1 + Var\, X_2 + 2\, Cov(X_1, X_2)$ (1)

$$var\,(X_1+X_2) = E\left[\left\{(X_1+X_2) - (\bar{X}_1+\bar{X}_2)\right\}^2\right]$$

$$= E\left[\left\{(X_1-\bar{X}_1) + (X_2-\bar{X}_2)\right\}^2\right]$$

$$= E\left[(X_1-\bar{X}_1)^2\right] + E\left[(X_2-\bar{X}_2)^2\right] + 2\, E\left[(X_1-\bar{X}_1)(X_2-\bar{X}_2)\right]$$

$$= Var\, X_1 + Var\, X_2 + 2\, Cov(X_1, X_2)$$ (2)

$$Var(X_1+X_2) = 1+1+2\times\frac{1}{4} = \frac{5}{2}$$ (3)

Q2 For x $(0<x<1)$, $E(Y|x) = \frac{1}{2}(x+1)$,

the mid-point of the interval $[x, 1]$ (1) (1)

Therefore $E[Y|X] = \frac{1}{2}(X+1)$ (2)

$$E[Y] = E[\,E[Y|X]] = \frac{1}{2}E[X+1] = \frac{1}{2}E(X) + \frac{1}{2}$$

$$= \frac{1}{2}\left(\frac{1}{2}+1\right) = \frac{3}{4}$$ (3)

Q3. $E[X] = (-1)\times\frac{3}{8} + 1\times\frac{5}{8} = \frac{1}{4}$

$E[X^2] = 1\times\frac{3}{8} + 1\times\frac{5}{8} = 1$, $\sigma_X^2 = 1-\left(\frac{1}{4}\right)^2 = \frac{15}{16}$ (1)

$E[Y] = 0\times\frac{4}{8} + 1\times\frac{4}{8} = \frac{1}{2}$,

$E[Y^2] = 0\times\frac{4}{8} + 1\times\frac{4}{8} = \frac{1}{2}$, $\sigma_Y^2 = \frac{1}{2}-\frac{1}{4} = \frac{1}{4}$ (2)

$E[XY] = 0\times(-1)\times\frac{1}{8} + 0\times1\times\frac{3}{8} + 1\times(-1)\times\frac{2}{8} + 1\times1\times\frac{2}{8} = 0$

$cov(XY) = 0 - \frac{1}{4}\times\frac{1}{2} = -\frac{1}{8}$ (3)

$$\rho_{XY} = \frac{-\frac{1}{8}}{\sqrt{\frac{15}{16}\times\frac{1}{4}}} = -\frac{1}{\sqrt{15}}.$$ (4)

Q4.

$$\mathrm{cov}(X,Y) = \mathrm{cov}(U+V, U-V) = E[U^2 - V^2] - E[U+V]E[U-V]$$
$$= 0$$

X and Y are uncorrelated

Q5

$$E[x^l y^m] = \int_0^1\int_0^1 x^l y^m (x+y)\,dx\,dy$$
$$= \int_0^1\int_0^1 x^{l+1} y^m\,dx\,dy + \int_0^1\int_0^1 x^l y^{m+1}\,dx\,dy$$
$$= \frac{1}{(l+2)(m+1)} + \frac{1}{(l+1)(m+2)} \qquad (1)$$

We have

$$E[X] = E[Y] = \frac{7}{12} \qquad (2)$$
$$E[X^2] = E[Y^2] = \frac{5}{12}$$
$$\mathrm{Var}\,X = \mathrm{Var}\,Y = \frac{5}{12} - \frac{49}{144} = \frac{11}{144}$$
$$\mathrm{cov}(X,Y) = E[XY] - E(X)E(Y) = \frac{1}{3} - \frac{49}{144} = -\frac{1}{144}$$
$$\rho_{XY} = \frac{-\frac{1}{144}}{\frac{11}{144}} = -\frac{1}{11}. \qquad (3)$$